

Exercise 18.6

- ① Determine whether the following functions are increasing or decreasing or stationary at specified point.

① a) $f(x) = 2x^3 - x^2 + 5$ at $x=1$, $x=3$ and $x=0$

→ Soln,

$$f'(x) = 6x^2 - 2x$$

$$f'(1) = 6 \times 1^2 - 2 \times 1 = 6 - 2 = 4 > 0 \text{ (Increasing)}$$

$$f'(3) = 6 \times 3^2 - 2 \times 3 = 54 - 6 = 48 > 0 \text{ (Increasing)}$$

$$f'(0) = 6 \times 0^2 - 2 \times 0 = 0 \text{ (stationary)}$$

∴ The function is increasing at $x=1$ and $x=3$ and stationary at $x=0$.

② b) $f(x) = x^2 - x + 1$ at $x = \frac{1}{2}$, $x=0$

→ Soln,

$$f'(x) = 2x - 1$$

$$f'\left(\frac{1}{2}\right) = 2 \times \frac{1}{2} - 1 = 1 - 1 = 0 \text{ (Stationary)}$$

$$f'(0) = 2 \times 0 - 1 = -1 < 0 \text{ (Decreasing)}$$

∴ The function is decreasing at $x=0$ and stationary at $x = \frac{1}{2}$.

② Find the intervals on which the following functions are increasing or decreasing.

③ a) $f(x) = 3x^2 - 6x + 5$

→ Soln,

$$f'(x) = 6x - 6 \Rightarrow 6(x-1)$$

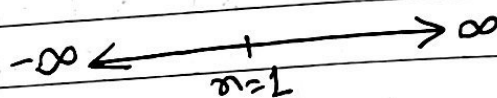
For turning point, $f'(x)$ must be zero. So,

$$6x - 6 = 0$$

$$\text{or, } 6x = 6$$

$$x = \frac{6}{6}$$

$$\therefore x = 1$$



The turning point $x=1$ divides the real line into two intervals $(-\infty, 1)$ and $(1, \infty)$

Now,

we observe the sign of $f'(x)$ in each interval.

Intervals	Sign of $f'(x) = 6(x-1)$	Remarks
$(-\infty, 1)$	-ve	decreasing
$(1, \infty)$	+ve	increasing

The function is increasing on $(1, \infty)$ and decreasing on $(-\infty, 1)$.

④ b) $f(x) = x^2 - 2x + 10$

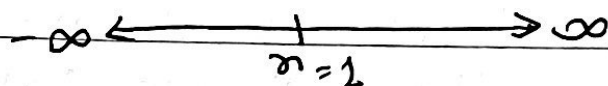
→ Soln,

$$f'(x) = 2x - 2 \Rightarrow 2(x-1)$$

For turning point, $f'(x)$ must be 0, so.

$$2(x-1) = 0$$

or $x = 1$



The turning point $x=1$ divides the real line into two intervals $(-\infty, 1)$ and $(1, \infty)$ Now,

We observe the sign of $f'(x)$ in each interval.

Intervals	Sign of $f'(x) = 2(x-1)$	Remarks
$(-\infty, 1)$	-ve	decreasing
$(1, \infty)$	+ve	increasing

∴ The function is increasing on $(1, \infty)$ and decreasing on $(-\infty, 1)$.

⑤ c) $f(x) = -2x^3 - 9x^2 - 12x + 1$

→ Soln,

$$f'(x) = -6x^2 - 18x - 12$$

For turning point, $f'(x)$ must be 0, so

$$-6x^2 - 18x - 12 = 0$$

$$\Rightarrow 6x^2 + 18x + 12 = 0$$

$$\Rightarrow x^2 + 3x + 2 = 0$$

$$\Rightarrow x^2 + 2x + x + 2 = 0$$

$$\Rightarrow x(x+2) + 1(x+2) = 0$$

$$\Rightarrow (x+1)(x+2)=0$$

Sither,

$$x+1=0$$

$$\text{or, } x=-1$$

$$x+2=0$$

$$x=-2$$



The turning point $x=-2$ and $x=-1$ divides the real line into three intervals $(-\infty, -2)$, $(-2, -1)$ and $(-1, \infty)$

Now, we observe the sign of $f'(x)$ in each interval.

Intervals	Sign of $f'(x) = (x+1)(x+2)$	Remarks
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$(-\infty, -2)$	+ve	Increasing
$(-2, -1)$	-ve	Decreasing
$(-1, \infty)$	+ve	Increasing

$\therefore$ The function is increasing on $(-\infty, -2) \cup (-1, \infty)$ and decreasing on $(-2, -1)$.

$$\textcircled{6} \text{ d) } f(x) = x + \frac{25}{x}$$

$$= x + 25x^{-1}$$

$$f'(x) = 1 + (-25)x^{-2}$$

$$= 1 - \frac{25}{x^2}$$

For turning point $f'(x)$ must be 0. so,

$$0 = 1 - \frac{25}{x^2}$$

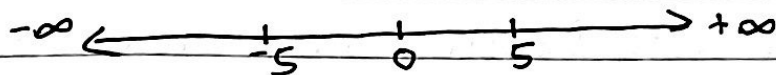
$$\text{or } \frac{25}{x^2} = 1$$

or, $x^2 = 25$

or, $x = \sqrt{25}$

$\therefore x = \pm 5$

$f'(x)$ is undefined when $x=0$ (critical point)



The turning point $x=(-5,5)$ and critical point $x=0$ ~~se~~ separates the intervals in four intervals.

$(-\infty, -5), (-5, 0), (0, 5), (5, \infty)$

We observe the sign of $f'(x)$ in each interval.

Interval	Sign of $f'(x) = 2 - 25/x^2$	Remarks
$(-\infty, -5)$	+ve	Increasing
$(-5, 0)$	-ve	decreasing
$(0, 5)$	-ve	decreasing
$(5, \infty)$	+ve	Increasing

$\therefore$ The function is increasing on $(-\infty, -5)$ and $(5, \infty)$ and decreasing on $(-5, 0)$ and $(0, 5)$.

3. Determine whether the following curves are concave downwards and ~~where~~ where they are concave ~~down~~ upwards.

⑦ a) $f(x) = 2x^3 - 6x^2 + 5$

→ Soln,

$$f'(x) = 6x^2 - 12x$$

$$f''(x) = 12x - 12$$

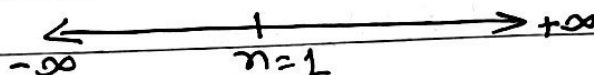
$$= 12(x-1)$$

The interval where function is concave downward or concave upward are separated by Point of inflection where $f''(x) = 0$

$$12(x-1) = 0$$

or, $x - 1 = 0$

$$\therefore x = 1$$



The point of inflection $x=1$ divides the real lines into two intervals $(-\infty, 1)$ and $(1, \infty)$.

We observe the sign of $f''(x)$ in each interval.

Interval	Sign of $f''(x) = 12(x-1)$	Remarks
$(-\infty, 1)$	-ve	Concave downward
$(1, \infty)$	+ve	Concave upward

$\therefore$ The function is concave downward on $(-\infty, 1)$ and upward on $(1, \infty)$ $\downarrow$

Q. b) $f(x) = x^3 - 6x^2 + 3x - 1$

→ Soln,

$$f'(x) = 3x^2 - 12x + 3$$

$$f''(x) = 6x - 12$$

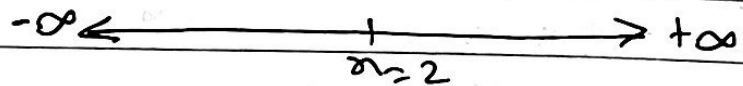
$$= 6(x - 2)$$

The interval where function is concave downward or concave upward are separated by point of inflection where $f''(x) = 0$.

$$0 = 6(x - 2)$$

$$x - 2 = 0$$

$$x = 2$$



The point of inflection $x=2$ divides the real lines into two intervals $(-\infty, 2)$ and $(2, \infty)$

We observe the sign of $f''(x)$ in each intervals.

Interval	Sign of $f''(x) = 6(x-2)$	Remarks
$(-\infty, 2)$	-ve	Concave downward
$(2, \infty)$	+ve	Upward

∴ The function is concave downward on $(-\infty, 2)$ and Upward on $(2, \infty)$.

Q) c) $f(x) = x^4 - 8x^3 + 18x^2 - 24$

→ Soln,

$$f'(x) = 4x^3 - 24x^2 + 36x$$

$$f''(x) = 12x^2 - 48x + 36$$

The interval where function is Concave downward ~~and~~ or concave upward are separated by point of inflection where $f''(x) = 0$

$$12x^2 - 48x + 36 = 0$$

or, $12(x^2 - 4x + 3) = 0$

or, $x^2 - 3x - x + 3 = 0$

or, $x(x-3) - 1(x-3) = 0$

or, $(x-1)(x-3) = 0$

Either,

$$x = 1, 3$$



The point of inflection $x=1, 3$ divides the real line into 3 intervals $(-\infty, 1)$, $(1, 3)$ and $(3, \infty)$

We observe the sign of $f''(x)$ in each interval.

Interval	Sign of $f''(x) = (x-1)(x-3)$	Remarks
$(-\infty, 1)$	+ve	Concave Upward
$(1, 3)$	-ve	Concave downward
$(3, \infty)$	+ve	Concave Upward

∴ The function is concave upward on $(-\infty, 1)$ and $(3, \infty)$ and concave downward on $(1, 3)$

4. Find the local maxima and local minima for the functions. Also, find the point of inflection (if any).

Q. a) $f(x) = 3x^2 - 6x + 4$

→ Soln,

$$f'(x) = 6x - 6$$

$$f''(x) = 6$$

For turning point,

$$f'(x) = 0$$

$$6x - 6 = 0$$

$$\text{or } 6x = 6$$

$$\therefore x = 1$$

at $x = 1$, $f''(1) = 6 > 0 \Rightarrow$ Concave upward

U → minimum

$$\text{min. Value} = f(1)$$

$$= 3x^2 - 6x + 4$$

$$= 3 - 6 + 4$$

$$= 1$$

$\therefore$ Minimum Value is 1 at $x = 1$.

For point of inflection $f''(x) = 0$

$$6 = 0 \text{ (Impossible)}$$

$\therefore$ The function has no point of inflection.

⑪ b) $f(x) = 4x^3 - 15x^2 + 12x + 7$

→ Soln,

$$f'(x) = 12x^2 - 30x + 12$$

$$f''(x) = 24x - 30$$

For turning point,

At $x = \frac{1}{2}$

$$f'(x) = 0$$

$$12x^2 - 30x + 12 = 0$$

or, ~~6x^2~~ $6(2x^2 - 5x + 2) = 0$

$$= 12 - 30$$

or, $2x^2 - 4x - x + 2 = 0$

$$= -18 < 0 \text{ (down-}$$

or, $2x(x-2) - 1(x-2) = 0$

ward)

or, $(2x-1)(x-2) = 0$

↖ min. Value

Either,

$$2x-1=0$$

$$x-2=0$$

$$x = \frac{1}{2}$$

$$x = 2$$

At $x = 2$

$$f''(x) = 24 \times 2 - 30$$

$$= 48 - 30 = 18 > 0$$

Upward

↘ min. Value

Min. Value = $4 \times 2^3 - 15 \times 2^2 + 12 \times 2 + 7$

$$= 32 - 60 + 24 + 7$$

$$= 3$$

Max Value = $4 \times \left(\frac{1}{2}\right)^3 - 15 \times \left(\frac{1}{2}\right)^2 + 12 \times \frac{1}{2} + 7$

$$= \frac{1}{2} - 15 \times \frac{1}{4} + 6 + 7$$

$$= \frac{39}{4}$$

∴ Minimum value is 3 at $x = 2$ and maximum value is $\frac{39}{4}$ at $x = \frac{1}{2}$

For point of inflection,

$$f''(x) = 0$$

$$24x - 30 = 0$$

$$x = \frac{30}{24}$$

$$x = \frac{5}{4}$$

12) c) $f(x) = 4x^3 - 6x^2 - 9x + 1$

→ Soln,

$$f'(x) = 12x^2 - 12x - 9$$

$$f''(x) = 24x - 12$$

For turning point

At $x = \frac{3}{2}$

$$f'(x) = 0$$

$$12x^2 - 12x - 9 = 0$$

$$12x^2 - 18x + 6x - 9 = 0$$

$$6x(2x - 3) + 3(2x - 3) = 0$$

$$(6x + 3)(2x - 3) = 0$$

$$f''\left(\frac{3}{2}\right) = 24 \times \frac{3}{2} - 12$$

$$= 36 - 12$$

$$= 24 > 0 \text{ (Upward)}$$

Either,

$$2x - 3 = 0$$

$$x = \frac{3}{2}$$

$$6x + 3 = 0$$

$$x = -\frac{1}{2}$$

U
Min.
Value.

At $x = -\frac{1}{2}$

$$\begin{aligned} \text{Min. Value} &= 4 \times \left(\frac{3}{2}\right)^3 - 6 \times \left(\frac{3}{2}\right)^2 \\ &\quad - 9 \times \frac{3}{2} + 1 \end{aligned}$$

$$= 4 \times \frac{27}{8} - 6 \times \frac{9}{4} - \frac{27}{2} + 1$$

$$= -\frac{25}{2}$$

$$f''\left(-\frac{1}{2}\right) = 24 \times -\frac{1}{2} - 12$$

$$= -12 - 12$$

$$= -24 \text{ (downward)}$$

∩
Max.
Value

$$\begin{aligned}
 \text{Max. Value} &= 4x\left(-\frac{1}{2}\right)^3 - 6x\left(-\frac{1}{2}\right)^2 - 9x\left(-\frac{1}{2}\right) + 1 \\
 &= 4x \cdot \frac{-1}{8} - 6x \cdot \frac{1}{4} + \frac{9}{2} + 1 \\
 &= -\frac{1}{2} - \frac{3}{2} + \frac{9}{2} + 1 \\
 &= \frac{7}{2}
 \end{aligned}$$

$\therefore$ Maximum value is $\frac{7}{2}$ at $x = -\frac{1}{2}$ and

minimum value is $-\frac{25}{2}$ at $x = \frac{3}{2}$ ✓

For point of inflection,

$$f''(x) = 0$$

$$24x - 12 = 0$$

$$\text{or } 24x = 12$$

$$\text{or } x = \frac{12}{24} = \frac{1}{2}$$

$\therefore$ point of inflection is at $\frac{1}{2}$ ✓

13) d) $f(x) = x + \frac{1}{x} + 2$

$$= x + x^{-1} + 2$$

$$f'(x) = 1 - 1x^{-2}$$

$$f''(x) = 2x^{-3} = \frac{2}{x^3}$$

For turning point,

$$f'(x) = 0$$

$$1 - \frac{1}{x^2} = 0$$

$$\text{or, } \frac{1}{x^2} = 1$$

$$\text{or, } x^2 = 1$$

$$\therefore x = \pm 1$$

At $x = +1$

$$f''(1) = \frac{2}{1^3} = 2 > 0 \quad \downarrow$$

min Value.

At $x = -1$

$$f''(-1) = \frac{2}{(-1)^3} = \frac{2}{-1} = -2 < 0$$

$\nearrow$ max. value

$$\text{Max Value} = -1 + \frac{1}{-1} + 2$$

$$= -1 - 1 + 2$$

$$= 0$$

$$\text{Min. Value} = 1 + \frac{1}{1} + 2 = 1 + 1 + 2 = 4$$

$\therefore$ Maximum value is 0 at $x = -1$ and minimum value is 4 at $x = 1$.

For point of inflection,

$$f''(x) = 0$$

$$\frac{2}{x^3} = 0$$

$$2 = 0 \text{ (Impossible)}$$

$\therefore$ The function has no point of inflection.

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Q 5. For the function $y = x + \frac{1}{x}$, show that maximum value is less than the minimum value.

→ Soln,

$$y = x + \frac{1}{x} = x + x^{-1}$$

$$\frac{dy}{dx} = 1 - x^{-2}$$

$$\frac{d^2y}{dx^2} = 2x^{-3} = \frac{2}{x^3}$$

For turning point,

$$\frac{dy}{dx} = 0$$

$$\text{or, } 1 - \frac{1}{x^2} = 0$$

$$\text{or, } 1 = \frac{1}{x^2}$$

$$\text{or, } x^2 = 1$$
$$\therefore x = \pm 1$$

At $x = 1$

$$\frac{d^2y}{dx^2} = f''(1) = \frac{2}{1^3} = 2 > 0$$

(Upward) $\cup$ min Value.

At $x = -1$

$$\frac{d^2y}{dx^2} = f''(-1) = \frac{2}{(-1)^3} = -2 < 0$$

(downward) $\cap$ max Value

$$\text{Max. Value} = -1 + \frac{1}{-1} = -1 + -1 = -2$$

$$\text{Min. Value} = 1 + \frac{1}{1} = 1 + 1 = 2$$

Here, min value is greater than max value.

$$\text{i.e. } \text{Min} - \text{Max} = 2 - (-2) = 4 \text{ (+ve)}$$

6) Find the absolute maximum and the absolute minimum values of the following function on the given intervals.

Q) a) $f(x) = 3x^2 - 6x + 4$ on $[-1, 2]$

→ Soln,

$$f'(x) = 6x - 6$$

For turning point

$$f'(x) = 0$$

$$6x - 6 = 0$$

$$6x = 6$$

$$x = 1$$

$$\therefore x = 1$$

At end point

$$f(-1) = 3 \times (-1)^2 - 6(-1) + 4$$

$$= 3 + 6 + 4$$

$$= 13 \text{ (Maximum)}$$

At turning point

$$f(1) = 3 \times 1^2 - 6 \times 1 + 4$$

$$= 3 - 6 + 4$$

$$= 1 \text{ (Minimum)}$$

$\therefore$ Absolute maximum is 13 at -1 and

Absolute minimum is 1 at 1

16) b) $f(x) = 2x^3 - 9x^2 + 12x + 20$ on $[-1, 5]$

→ Soln,

$$f'(x) = 6x^2 - 18x + 12$$

For turning point,

$$f'(x) = 0$$

$$\text{or, } 6x^2 - 18x + 12 = 0$$

$$\text{or, } 6(x^2 - 3x + 2) = 0$$

$$\text{or, } x^2 - 2x - x + 2 = 0$$

$$\text{or, } x(x-2) - 1(x-2) = 0$$

$$\text{or, } (x-1)(x-2) = 0$$

Either,

$$x = 1, 2$$

Now, evaluating the values of $f(x)$ at the stationary points and end points of intervals, we have,

$$f(1) = 2 \times 1^3 - 9 \times 1^2 + 12 \times 1 + 20$$

$$= 2 - 9 + 12 + 20$$

$$= 25 \text{ (maximum)}$$

$$f(2) = 2 \times 2^3 - 9 \times 2^2 + 12 \times 2 + 20$$

$$= 16 - 36 + 24 + 20$$

$$= 24 \text{ (minimum)}$$

At end point

$$f(-1) = 2 \times (-1)^3 - 9 \times (-1)^2 + 12 \times (-1) + 20$$

$$= -2 - 9 - 12 + 20$$

$$= -3 \text{ (minimum)}$$

$$f(5) = 2 \times 5^3 - 9 \times 5^2 + 12 \times 5 + 20$$

$$= 250 - 225 + 60 + 20$$

$$= 105 \text{ (maximum)}$$

$\therefore$ Absolute maximum is 105 at 5 and
Absolute minimum is -3 at -1.

⑦) $f(x) = 2x^3 - 15x^2 + 36x + 10$ on $[1, 4]$

→ Soln,

$$f'(x) = 6x^2 - 30x + 36$$

For turning point,

$$f'(x) = 0$$

$$\cancel{2x^3 - 15x^2 + 36x} \quad 6x^2 - 30x + 36 = 0$$

$$6(x^2 - 5x + 6) = 0$$

$$x^2 - 5x + 6 = 0$$

$$\text{or, } x^2 - 3x - 2x + 6 = 0$$

$$\text{or, } x(x-3) - 2(x-3) = 0$$

$$\text{or, } (x-2)(x-3) = 0$$

Either,

$$x = 2, 3$$

Now, evaluating the values of $f(x)$ at the stationary points and end points of intervals.

We have,

$$\begin{aligned} f(3) &= 2 \times 3^3 - 15 \times 3^2 + 36 \times 3 + 10 \\ &= 37 \end{aligned}$$

$$\begin{aligned} f(2) &= 2 \times 2^3 - 15 \times 2^2 + 36 \times 2 + 10 \\ &= 38 \end{aligned}$$

$$\begin{aligned} f(1) &= 2 \times 1^3 - 15 \times 1^2 + 36 \times 1 + 10 \\ &= 33 \text{ (Minimum)} \end{aligned}$$

$$\begin{aligned} f(4) &= 2 \times 4^3 - 15 \times 4^2 + 36 \times 4 + 10 \\ &= 42 \text{ (Maximum)} \end{aligned}$$

7. Show that the following functions have neither maximum nor minimum value.

(18) a) $f(x) = x^3 - 3x^2 + 9x + 5$

→ Soln,

$$f'(x) = 3x^2 - 6x + 9$$

$$f''(x) = 6x - 6$$

For turning point,

$$f'(x) = 0$$

$$3x^2 - 6x + 9 = 0$$

$$3(x^2 - 2x + 3) = 0$$

$$x^2 - 2x + 3 = 0$$

$$a = 1, b = -2, c = 3$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{2 \pm \sqrt{2^2 - 4 \times 1 \times 3}}{2 \times 1}$$

$$= \frac{2 \pm \sqrt{4 - 12}}{2}$$

$$= \frac{2 \pm \sqrt{-8}}{2}$$

Which is imaginary.

So the function have neither maximum nor minimum value.

(19) b) $f(x) = x^3 - 3x^2 + 3x + 4$

→ Soln,

$$f'(x) = 3x^2 - 6x + 3$$

for turning point,

$$f'(x) = 0$$

$$3x^2 - 6x + 3 = 0$$

$$\text{or, } 3x^2 - 3x - 3x + 3 = 0$$

$$\text{or, } 3x(x-1) - 3(x-1) = 0$$

$$\text{or, } (3x-3)(x-1) = 0$$

Either,

$$3x-3=0 \quad x-1=0$$

$$x=1 \quad x=1$$

$$f''(x) = 6x - 6$$

$$f''(1) = 6 \times 1 - 6$$

$$= 6 - 6$$

$$= 0$$

The function is neither concave upward nor downward hence it has neither maximum nor minimum.

8.

20. a) Find maximum value of x^2y such that

$$x+y=10$$

→ Soln,

$$x+y=10$$

$$y=10-x \rightarrow (i)$$

$$\text{let, } x^2y = A$$

$$\frac{dA}{dx} \Rightarrow A = x^2(10-x)$$

$$A = 10x^2 - x^3$$

$$\frac{dA}{dx} = 20x - 3x^2$$

$$\frac{d^2A}{dx^2} = 20 - 6x$$

For turning point,

$$\frac{dA}{dx} = 0$$

$$\text{At } x = \frac{20}{3}$$

$$20x - 3x^2 = 0$$

$$\text{or, } 3x^2 - 20x = 0$$

$$\text{or, } x(3x - 20) = 0$$

Either,

$$x = 0, \quad 3x - 20 = 0$$

$$x = \frac{20}{3}$$

$$\frac{d^2A}{dx^2} = 20 - 6x$$

$$= 20 - 6 \times \frac{20}{3}$$

$$= 20 - 40$$

$$= -20 < 0$$

(downward)

↗ Max Value.

$$\text{Max. Value} = 10 \times \left(\frac{20}{3}\right)^2 - \left(\frac{20}{3}\right)^3$$

$$= 10 \times \frac{400}{9} - \frac{8000}{27}$$

$$= \frac{4000 \times 3 - 8000}{27}$$

$$= \frac{4000}{27}$$

$$\therefore \text{Maximum Value of } x^2y = \frac{4000}{27}$$

21) b) Find the minimum value of $x^2 + y^2$, where $x + y = 5$

→ Soln,

$$x + y = 5$$

$$x = 5 - y \rightarrow (i)$$

let,

$$x^2 + y^2 = A$$

$$\text{or, } (5 - y)^2 + y^2 = A$$

$$\text{or, } A = 25 - 10y + 2y^2$$

$$\text{or, } A = 2y^2 - 10y + 25$$

$$\frac{dA}{dn} = 4y - 10$$

$$\frac{d^2A}{dn^2} = 4$$

$$\text{At } y = \frac{5}{2}$$

For turning point,

$$\frac{dA}{dn} = 0$$

$$\frac{d^2A}{dn^2} = 4 > 0 \quad (\text{Upward})$$

$$4y - 10 = 0$$

$$4y = 10$$

 (Min value)

$$\text{or, } y = \frac{10}{4} = \frac{5}{2}$$

$$\text{Min Value} = 2 \times \left(\frac{5}{2}\right)^2 - 10 \times \frac{5}{2} + 25$$

$$= 2 \times \frac{25}{4} - 5 \times 5 + 25$$

$$= \frac{25}{2}$$

$\therefore$ Minimum value of $x^2 + y^2$ is $\frac{25}{2}$

- ② c) Amongst all pairs of positive numbers with product 64, find those whose sum is the least.

→ Soln,

let two positive number be x and y .

$$x \cdot y = 64$$

$$x = \frac{64}{y} \rightarrow (i)$$

$$\text{let, } S = x + y$$

$$S = \frac{64}{y} + y$$

$$S = \frac{64 + y^2}{y}$$

$$S = 64y^{-1} + y$$

$$\frac{dS}{dy} = -64y^{-2} + 1$$

$$\frac{d^2S}{dy^2} = \frac{128}{y^3} + 1$$

For turning point,

$$\frac{dS}{dy} = 0$$

$$-\frac{64}{y^2} + 1 = 0$$

$$\text{or, } \frac{64}{y^2} = 1$$

$$\text{or, } y^2 = 64$$

$$\therefore y = \pm 8$$

Now,

$$S = x + y$$

$$16 = x + 8$$

$$x = 16 - 8$$

$$\therefore x = 8$$

$$\text{At } y = 8$$

$$\frac{d^2S}{dy^2} = \frac{128}{8} = 16 > 0$$

(Upwards)

↪ minimum

$$\text{minimum } S = \frac{64}{8} + 8$$

$$S = 8 + 8$$

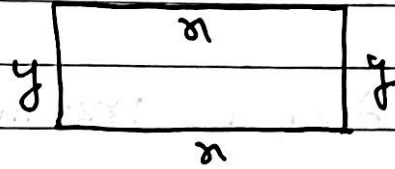
$$S = 16 \text{ (minimum)}$$

Those two positive numbers are 8. and 8.

9. a) Show that the rectangle of largest possible area for a given perimeter is square.

→ Soln,

let x and y be length and breadth of a rectangle with given perimeter P (constant)



$$P = 2x + 2y$$

$$\text{or, } x + y = \frac{P}{2}$$

$$\text{or, } y = \frac{P}{2} - x \rightarrow (i)$$

$$\text{Area (A)} = x \times y$$

$$A = x \left(\frac{P}{2} - x \right)$$

$$A = \frac{P}{2}x - x^2$$

$$\frac{dA}{dx} = \frac{P}{2} - 2x$$

$$\frac{d^2A}{dx^2} = -2$$

For turning point,

$$\frac{dA}{dx} = 0$$

$$\frac{P}{2} - 2x = 0$$

$$\text{or, } 2x = \frac{P}{2} \Rightarrow x = \frac{P}{4}$$

$$\text{at } x = \frac{p}{4}$$

$$\frac{d^2A}{dx^2} = -2 < 0 \text{ (concave downward)}$$

↖ Max. value

Area is maximum at $x = \frac{p}{4}$

$$\text{From (i)} \quad y = \frac{p}{2} - \frac{p}{4}$$

$$y = \frac{p}{4}$$

$$\therefore x = y = \frac{p}{4}$$

Hence the rectangle is square for maximum area.

- (24) b. A man wishes to fence a rectangular garden with 256 m fencing material. Find the maximum area he can enclosed.

→ Soln,

let length be x and breadth be y .

$$2(x+y) = 256$$

$$x+y = \frac{256}{2}$$

$$y = 128 - x \rightarrow \text{(i)}$$

$$\text{let, } A = x \cdot y$$

$$\text{or, } A = x(128 - x)$$

$$\text{or, } A = 128x - x^2$$

$$\frac{dA}{dx} = 128 - 2x$$

$$\frac{d^2A}{dn^2} = -2 < 0 \text{ (concave downward)}$$

Turning point,

$$\frac{dA}{dn} = 0$$

$$128 - 2n = 0$$

$$\text{or, } 2n = 128$$

$$\text{or, } n = \frac{128}{2}$$

$$\therefore n = 64$$

At $n = 64$

$$\frac{d^2A}{dn^2} = -2 < 0 \text{ (downward)}$$

max
Value

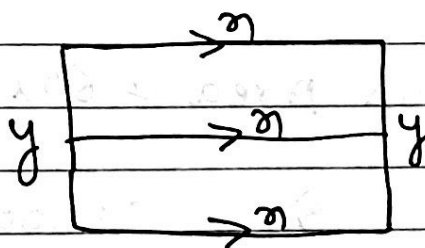
$$\begin{aligned} \text{man. Area} &= 128 \times 64 - (64)^2 \\ &= 8192 - 4096 \\ &= 4096 \end{aligned}$$

$\therefore$ Maximum Area he can enclosed is 4096 m^2 .

- 25) c) A garden having 120 m of fencing wishes to enclose a rectangular plot of land and also erect a fence across the land parallel to two of the sides. Find the maximum area he can enclose.

→ Soln,

let x and y be length and breadth of the land enclosed



by a fence of length

120 m and also to erect a fence across the land parallel to two of the sides.

By question,

$$3x + 2y = 120$$

$$\text{or, } 2y = 120 - 3x$$

$$\text{or, } y = \frac{120 - 3x}{2} \rightarrow (i)$$

let A be Area of the land then,

$$A = x \times y \\ = x \left(60 - \frac{3x}{2} \right)$$

$$= 60x - \frac{3}{2}x^2$$

$$\frac{dA}{dx} = 60 - \frac{3}{2} \times 2x$$

$$= 60 - 3x$$

$$\frac{d^2A}{dx^2} = -3$$

$$\text{At } x=20$$

$$\frac{d^2A}{dx^2} = -3 < 0 \text{ (-ve)}$$

(downward)

→ max. Values.

For turning point,

$$\frac{dA}{dx} = 0$$

$$60 - 3x = 0$$

$$3x = 60$$

$$x = 20$$

Area becomes maximum when, $x=20$

$$\text{Max Area} = 60 \times 20 - \frac{3}{2} \times 20^2$$

$$= 1200 - 600$$

$$= 600 \text{ m}^2$$